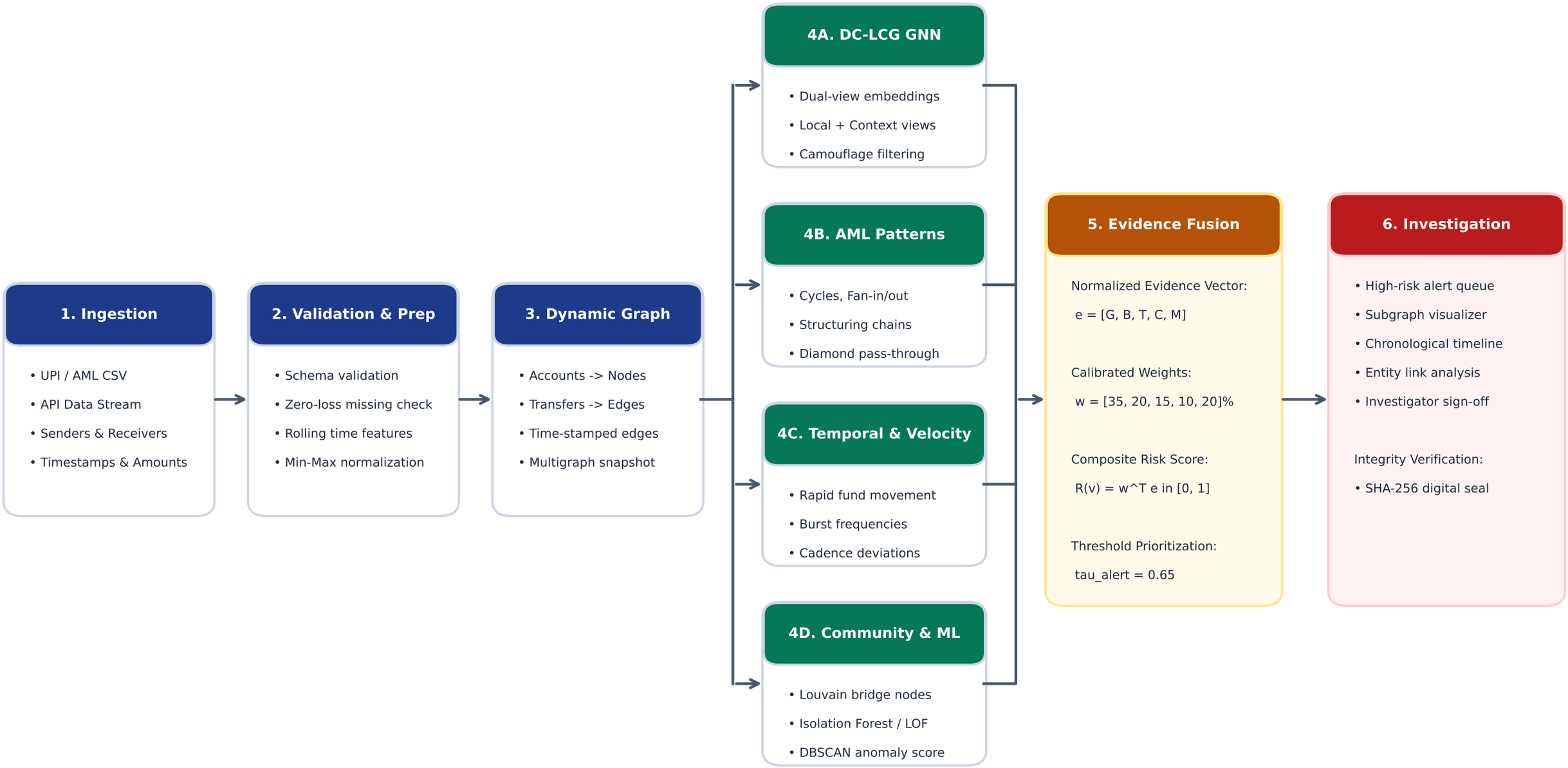
1. Ingestion
• UPI / AML CSV
• API Data Stream
• Senders & Receivers
• Timestamps & Amounts
2. Validation & Prep
• Schema validation
• Zero-loss missing check
• Rolling time features
• Min-Max normalization
3. Dynamic Graph
• Accounts -> Nodes
• Transfers -> Edges
• Time-stamped edges
• Multigraph snapshot
4A. DC-LCG GNN
• Dual-view embeddings
• Local + Context views
• Camouflage filtering
4B. AML Patterns
• Cycles, Fan-in/out
• Structuring chains
• Diamond pass-through
4C. Temporal & Velocity
• Rapid fund movement
• Burst frequencies
• Cadence deviations
4D. Community & ML
• Louvain bridge nodes
• Isolation Forest / LOF
• DBSCAN anomaly score
5. Evidence Fusion
Normalized Evidence Vector:
e = [G, B, T, C, M]
Calibrated Weights:
w = [35, 20, 15, 10, 20]%
Composite Risk Score:
R(v) = w^T e in [0, 1]
Threshold Prioritization:
tau_alert = 0.65
6. Investigation
• High-risk alert queue
• Subgraph visualizer
• Chronological timeline
• Entity link analysis
• Investigator sign-off
Integrity Verification:
• SHA-256 digital seal